

Optimization II

Equality Constraints

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Lagrange multiplier method.

- 1 2 variables - 1 constraint
- 2 Multiple variables - multiple constraints
- 3 Some research applications

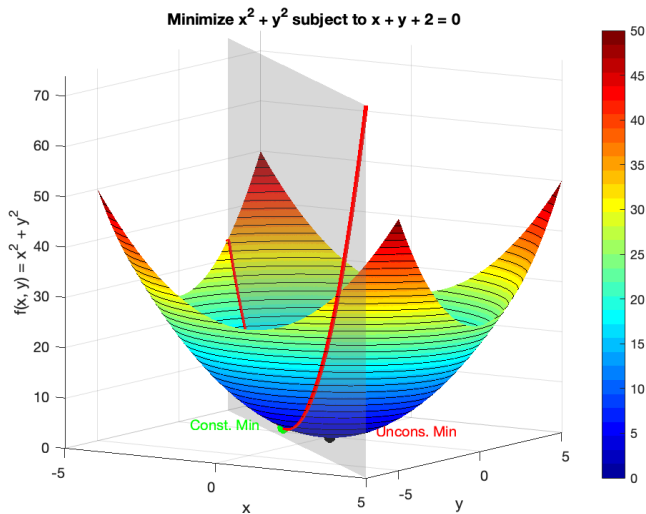
2 variables - 1 constraint

Unconstrained optimization

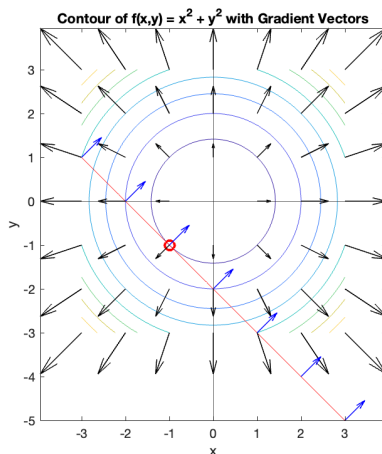
$$\min f(x, y) := x^2 + y^2$$

Equality-constrained optimization

$$\begin{aligned} \min f(x, y) &:= x^2 + y^2 \\ \text{s.t. } h(x, y) &:= x + y + 2 = 0 \end{aligned}$$



At the critical point



- The slope of f and h must be equal.
- The gradient vectors of f and h are parallel.

Slope of f at (x^*, y^*)

$$y'(x) = -\frac{\partial f / \partial x}{\partial f / \partial y}(x^*, y^*)$$

Slope of h at (x^*, y^*)

$$y'(x) = -\frac{\partial h / \partial x}{\partial h / \partial y}(x^*, y^*)$$

Equal slopes are equivalent to

$$\frac{\partial f / \partial x}{\partial h / \partial x} = \frac{\partial f / \partial y}{\partial h / \partial y} = \lambda$$

which is equivalent to

$$\frac{\partial f}{\partial x}(x^*, y^*) - \lambda \frac{\partial h}{\partial x}(x^*, y^*) = 0,$$

$$\frac{\partial f}{\partial y}(x^*, y^*) - \lambda \frac{\partial h}{\partial y}(x^*, y^*) = 0,$$

Expressed in matrix form

$$\nabla f(x^*, y^*) - \lambda \nabla h(x^*, y^*) = 0.$$

We can transform the original objective function into a Lagrangian function of a 3-variable unconstrained optimization problem (*)

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda[c - h(x, y)]$$

FOC

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial y} = \frac{\partial \mathcal{L}}{\partial \lambda} = 0.$$

Note:

- $\partial h / \partial x, \partial h / \partial y$ must be $\neq 0$.
- λ is called the Lagrange multiplier.
- (*) λ is a special variable, not like x, y .

Example 1

Find extrema

$$\begin{aligned} f(x, y) &:= x^2 + y^2, \\ \text{s.t. } x + y &= -2. \end{aligned}$$

- 1 Write the Lagrangian function

$$\mathcal{L} = x^2 + y^2 + \lambda(-2 - x - y)$$

- 2 FOC (First-Order Condition)

$$\mathcal{L}_\lambda = -2 - x - y = 0,$$

$$\mathcal{L}_x = 2x - \lambda = 0,$$

$$\mathcal{L}_y = 2y - \lambda = 0$$

- 3 Solution: $x^* = y^* = -1$ and $\lambda = -2$

Example 2



Find extrema

$$\begin{aligned} f(x, y) &:= xy, \\ \text{s.t. } x + y &= 6. \end{aligned}$$

- 1 Write the Lagrangian function

$$\mathcal{L} = xy + \lambda(6 - x - y)$$

- 2 FOC

$$\begin{aligned} \mathcal{L}_\lambda &= 6 - x - y = 0, \\ \mathcal{L}_x &= y - \lambda = 0, \\ \mathcal{L}_y &= x - \lambda = 0 \end{aligned}$$

- 3 Solution: $x^* = y^* = 3$ and $\lambda = 3$

Find extrema

$$\begin{aligned} f(x, y) &:= x^2 y, \\ \text{s.t. } 2x^2 + y^2 &= 3. \end{aligned}$$

- 1 Write the Lagrangian function

$$\mathcal{L} = x^2 y + \lambda(3 - 2x^2 - y^2)$$

- 2 FOC:

$$(x) : 2xy - 4\lambda x = 0,$$

$$(y) : x^2 - 2\lambda y = 0,$$

$$(\lambda) : 2x^2 + y^2 = 3$$

- 3 We obtain 6 solutions (x^*, y^*, λ^*) : $(0, \pm\sqrt{3}, 0), (\pm 1, 1, 1/2), (\pm 1, -1, -1/2)$.

Find extrema:

- ① $f(x, y) := x^2 + y^2$ s.t. $x + 4y = 2$
- ② $f(x, y) = x(y + 4)$ s.t. $x + y = 8$
- ③ $f(x, y) = x - 3y - xy$ s.t. $x + y = 6$
- ④ $f(x, y) = 7 - y + x^2$ s.t. $x + y = 0$
- ⑤ $f(x, y) = \ln x + \gamma \ln(1 - y)$ s.t. $ax + c = yw$ with $\gamma, a, c, w > 0, x > 0, 0 < y < 1$
- ⑥ $f(x, y) = 5x - 3y$ s.t. $x^2 + y^2 = 136$

- FONC only tells us the location of critical points.
- To check whether a point is a min or max, we need to consider additional conditions.

Assume both functions f, h are twice differentiable. The critical point of

$$z = f(x, y) \quad \text{s.t.} \quad h(x, y) = c$$

is (x^*, y^*) satisfying the following first-order conditions:

$$\mathcal{L}'_x = \mathcal{L}'_y = \mathcal{L}'_\lambda = 0 \quad \text{at } (x^*, y^*, \lambda^*)$$

where the Lagrangian function is:

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda(c - h(x, y))$$

Consider the bordered Hessian

$$\overline{H} = \begin{pmatrix} 0 & \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} \\ \frac{\partial h}{\partial x} & \frac{\partial^2 \mathcal{L}}{\partial x^2} & \frac{\partial^2 \mathcal{L}}{\partial x \partial y} \\ \frac{\partial h}{\partial y} & \frac{\partial^2 \mathcal{L}}{\partial y \partial x} & \frac{\partial^2 \mathcal{L}}{\partial y^2} \end{pmatrix}$$

- If $\det(\overline{H}) > 0$ at (x^*, y^*, λ^*) , then it is a local max
- If $\det(\overline{H}) < 0$ at (x^*, y^*, λ^*) , then it is a local min.
- If $\det(\overline{H}) = 0$, then it is a saddle point.

- Due to the presence of constraint h , when considering second-order conditions, we can only consider changes that do not violate the constraint.
- In other words, we need to check the second derivative of the Lagrangian but restricted to the space that does not change the constraint. This is equivalent to checking the definiteness of the null space of $\nabla h(x^*)$. This is equivalent to checking the sign of

$$d^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) d$$

for all $d \in \mathbb{R}^n$ and $\nabla h(x^*)d = 0$. This information is concretized in the bordered Hessian.

- The idea of checking second derivatives in a 2-variable, 1-constraint problem: Using the constraint, we can express every point in the domain as $(x, \phi(x))$. Reduce the original problem from 2 variables to 1 variable, then find the second derivative. [▶▶ proof](#)

- Example 1: the solution is a min because

$$|\overline{H}| = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix} = -4 < 0$$

- Example 2: the solution is a max because

$$|\overline{H}| = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 2 > 0$$

- Example 3:

$$|\overline{H}| = \begin{vmatrix} 0 & 4x & 2y \\ 4x & 2y - 4\lambda & 2x \\ 2y & 2x & -2\lambda \end{vmatrix}$$

- $(\pm 1, -1, -1/2)$ and $(0, \sqrt{3}, 0)$: min
- $(\pm 1, 1, 1/2)$ and $(0, -\sqrt{3}, 0)$: max

Substitution Method (naive method)



If the constraint is a linear function, we should use the substitution method.

Consider ▶ Example 2

From the constraint, we have

$$y = 6 - x$$

The problem becomes

$$f(x) = x(6 - x)$$

FOC

$$f'(x) = 6 - 2x = 0$$

SOC

$$f''(x) = -2$$

So $x = 3$ is a max.

- Check whether the critical points are min or max [▶▶ exercises](#).
- For problems with linear constraints, use the substitution method.

The Lagrange multiplier indicates the change in the optimal function value when the constraint changes.

$$\max(\min)f(x, y) \text{ st. } h(x, y) = c$$

It can be proven that [» proof](#)

$$\frac{df^*(c)}{dc} = \lambda$$

Shadow price of the constraint. If Δc is a small change in the constraint, then

$$f^*(c + \Delta c) - f^*(c) \approx \lambda(c)dc$$

An individual lives for only two periods, spending on goods c_1 and c_2 in each period. This individual wants to maximize utility as follows

$$U(c_1, c_2) = c_1 c_2$$

Let $r > 0$ be the interest rate, $B > 0$ be the amount of money he has. The constraints are stated as follows:

$$\begin{aligned} c_1 + s &= B, \\ c_2 &= (1 + r)s \end{aligned}$$

- 1 Eliminate s to simplify the problem to 1 constraint.
- 2 Write the Lagrangian function.
- 3 Find optimal c_1^*, c_2^* .
- 4 Check whether they are truly optimal.

A factory wants to produce 30 cars as cheaply as possible. Using K capital and L labor produces $\sqrt{K} + L$ cars. Suppose the price per unit of capital and labor is 1 and 20, respectively.

- 1 Write the factory's problem.
- 2 Find optimal K and L .
- 3 Use SOC to verify optimality.
- 4 By how much does cost increase to produce 31 instead of 30 cars?

Multiple variables - multiple constraints

For each constraint, we introduce a separate Lagrange multiplier.
Objective function with n variables

$$z = f(x_1, x_2, \dots, x_n)$$

with k constraints

$$\mathcal{L} = f(x_1, \dots, x_n) + \sum_{i=1}^k \lambda_i (c_i - g_i(x_1, \dots, x_n))$$

First-order necessary condition to find critical points

$$\frac{\partial \mathcal{L}}{\partial x_i} = 0 \quad \text{for } i = 1, \dots, n$$

Example A



Find extrema

$$F(x, y, z) = x^2 + y^2 + z^2$$

constraints

$$g_1(x, y, z) = 3x + y + z = 5,$$

$$g_2(x, y, z) = x + y + z = 1$$

We find the critical point:

$$x^* = 2, \quad y^* = -1/2, \quad z^* = -1/2$$

along with the Lagrange multiplier values $\lambda_1 = 5/2$, $\lambda_2 = -7/2$

Second-Order Sufficient Conditions (SOC)

Case of n variables but only 1 constraint.

Bordered Hessian matrix

$$\overline{H} = \begin{pmatrix} 0 & g_1 & g_2 & g_3 & \cdots & g_n \\ g_1 & \mathcal{L}_{11} & \mathcal{L}_{12} & \mathcal{L}_{13} & \cdots & \mathcal{L}_{1n} \\ g_2 & \mathcal{L}_{21} & \mathcal{L}_{22} & \mathcal{L}_{23} & \cdots & \mathcal{L}_{2n} \\ g_3 & \mathcal{L}_{31} & \mathcal{L}_{32} & \mathcal{L}_{33} & \cdots & \mathcal{L}_{3n} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ g_n & \mathcal{L}_{n1} & \mathcal{L}_{n2} & \cdots & \cdots & \mathcal{L}_{nn} \end{pmatrix}$$

Check: $|\overline{H}_2|, |\overline{H}_3|, \dots, |\overline{H}_n|$ (note: $|\overline{H}_1|$ is always negative).

- If $|\overline{H}_2|, |\overline{H}_3|, \dots, |\overline{H}_n|$ (all bordered leading principal minors are negative, same sign as $(-1)^{k=1}$), then the extremum is a min.
- If $|\overline{H}_2| > 0, |\overline{H}_3| < 0, \dots, (-1)^n |\overline{H}_n| > 0$ (bordered leading principal minors alternate in sign with $\overline{H}_n$ having the same sign as $(-1)^n$), then the extremum is a max.

SOC for n variables and k constraints ($n > k$).

Further reading: FMMA chap.12, ME chap.18 & 19, MA pp.58,59.
Bordered Hessian matrix

$$\bar{H} = \left(\begin{array}{ccc|ccc} 0 & \dots & 0 & \frac{\partial g_1}{\partial x_1} & \dots & \frac{\partial g_1}{\partial x_n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \frac{\partial g_k}{\partial x_1} & \dots & \frac{\partial g_k}{\partial x_n} \\ \hline \frac{\partial g_1}{\partial x_1} & \dots & \frac{\partial g_k}{\partial x_1} & \frac{\partial^2 \mathcal{L}}{\partial x_1^2} & \dots & \frac{\partial^2 \mathcal{L}}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_1}{\partial x_n} & \dots & \frac{\partial g_k}{\partial x_n} & \frac{\partial^2 \mathcal{L}}{\partial x_n \partial x_1} & \dots & \frac{\partial^2 \mathcal{L}}{\partial x_n^2} \end{array} \right)$$

Check $(n - k)$ last leading bordered principal minors
Conditions

- (max) If signs alternate continuously with $|\bar{H}_n|$ having the same sign as $(-1)^n$.
- (min) If all have the same sign as $(-1)^k$.
- If neither of the 2 cases above $\rightarrow$ neither max nor min.

Return to [▶ Example A](#)

To determine whether this point is a min or max, set up the bordered Hessian

$$\bar{H} = \left(\begin{array}{cc|ccc} 0 & 0 & 3 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ \hline 3 & 1 & 2 & 0 & 0 \\ 1 & 1 & 0 & 2 & 0 \\ 1 & 1 & 0 & 0 & 2 \end{array} \right)$$

We have $n = 3, k = 2$, so we must check $n - k = 1$ last leading principal minors, meaning check the sign of the entire $\bar{H}$, i.e., $|\bar{H}_3|$

- If $|\bar{H}_3|$ has the same sign as $(-1)^{k=2} = 1$ (in other words, $(-1)^2 |\bar{H}_3| > 0$), then this is a min.
- If $|\bar{H}_3|$ has the same sign as $(-1)^{n=3} = -1$ (in other words, $(-1)^3 |\bar{H}_3| > 0$), then this is a max.

We compute $|\bar{H}_3| = 16 > 0$, which has the same sign as $(-1)^k = (-1)^2 > 0$, so we can conclude this is a MIN point.

(*) In this problem, we can use the 2 constraints to reduce the problem to 2 variables and 1 constraint. Try it.

Example B: [ME, 19.8, p465]



Find max

$$\max x^2 y^2 z^2 \text{ s.t. } x^2 + y^2 + z^2 = 3 \quad (1)$$

FOC gives the solution $x^2 = y^2 = z^2 = \lambda = 1$. Bordered Hessian

$$\bar{H} = \begin{pmatrix} 0 & 2x & 2y & 2z \\ 2x & 2y^2 z^2 - 2\lambda & 4xyz^2 & 4xy^2 z \\ 2y & 4xyz^2 & 2x^2 z^2 - 2\lambda & 4x^2 yz \\ 2z & 4xy^2 z & 4x^2 yz & 2x^2 y^2 - 2\lambda \end{pmatrix} \quad (2)$$

At the point $x = y = z = \lambda = 1$, we have:

$$\bar{H} = \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 0 & 4 & 4 \\ 2 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 \end{pmatrix} \quad (3)$$

Since the problem has $n = 3$ and $k = 1$, we need to check the signs of the last 2 leading minors.

Matrices $\bar{H}_2$ (red diagonal) and $\bar{H}_3$ (red+blue diagonal).

- $\det \bar{H}_2 = 32, \det \bar{H}_3 = -192$
- and $(-1)^{n-3} \bar{H}_3 > 0$

So we can conclude this critical point is a max.

Find optimal points and check sufficient conditions by setting up the bordered Hessian.

- ① $\max f(x, y) = x + y$ s.t. $x^2 + y^2 = 2$
- ② $\max f(x, y, z) = xyz$ s.t. $x + 2y + 3z = 6$
- ③ $\max f(x, y, z) = yz + xz$ s.t. $y^2 + z^2 = 1$ and $xz = 3$
- ④ $\min f(x, y, z) = x^2 + y^2 + z^2$ s.t. $x + 2y + z = 30$ and $2x - y - 3z = 10$
- ⑤ $\max f(x, y, z) = 10x^{1/2}y^{1/3}z^{1/4}$ s.t. $4x + 3y + 6z = 390$.
- ⑥ $\max(\min)f(x, y, z, w) = x^2 + y^2 + 2z^2 + 4w^2$ s.t. $x + y + z + w = 10$ and $2x - y + z + 3w = 5$

Applications

A representative agent consumes good x and chooses working time ℓ . The utility function is as follows:

$$U(x, \ell) = \alpha \ln x + \beta \ln(1 - \ell) \quad (4)$$

s.t.

$$px = w\ell + b$$

given that $0 \leq \ell \leq 1$ and $x, w, \alpha, \beta > 0$. Where p is the price of goods, w is wage, b is other income.

- ① Find (x^*, ℓ^*) to maximize utility. Check SOC.
- ② How does an increase in wage w affect optimal decisions?
- ③ Prove that $x^*(\mu p, \mu w, \mu b) = \mu^0 x^*(p, w, b) \forall \mu > 0$.
- ④ If b increases to $(1 + \gamma)b$ with γ being a small positive number, what happens to the new optimal utility? Compare the Lagrange multiplier with $\partial U / \partial b$.

A family is considering purchasing childcare services $s \in [0, \bar{s}]$ at price p , where $\bar{s}$ is the maximum childcare service. The family's utility function is:

$$\max_{c, n, s} \ln c + \delta \ln n \quad (5)$$

s.t.

$$\begin{aligned} c + \psi n + spn\phi &= w_m + w_f l_f, \\ l_f &= 1 - (1 - s)\phi n \end{aligned}$$

Where c, s, w_m, w_f, l_f are consumption, childcare services to purchase, husband's wage, wife's wage, and wife's working time. The time opportunity cost of childcare is $\phi \in (0, 1)$.

- ① Treating s as given, find optimal c^* and n^* .
- ② Substitute the values of c^* and n^* into function U , then compute the derivative $\partial U / \partial s$. Answer the question: when will this family purchase s ?
- ③ When s changes, how do n and l_f change?
- ④ Suppose $\psi < w_m\phi$, at what level of s does $\partial n / \partial w_f > 0$?

Appendix

EMEA : Essential mathematics for economic analysis (Sydsaeter & Hammond et. al, 6e 2021)

FMMA : Fundamental methods of mathematical economics (Chiang & Wainwright, 4e 2005)

FMEA : Further mathematics for economic analysis (Sydsaeter & Hammond et. al, 2e 2008)

ME : Mathematics for economists (Simon & Blume, 1994)

NMO : Numerical methods and optimization (Butenko & Pardalos, 2014)

ORAA : Operations Research: Applications and Algorithms (Winston, 2003)

MA : Microeconomic Analysis (Varian, 1992)

Related further reading: EMEA chap.14, FMMA chap.12, ME chap.18 & 19, MA pp.58,59.

Suppose we have

$$y = \phi(x) \quad (6)$$

notation:

$$\phi'(x) = \frac{\partial y}{\partial x} \quad (7)$$

Step 1: The constraint becomes

$$h(x, \phi(x)) = c$$

Total derivative of both sides at x^*

$$\frac{\partial h}{\partial x}(x^*) + \frac{\partial h}{\partial y}\phi'(x^*) = 0$$

We obtain

$$\phi'(x^*) = -\frac{\partial h / \partial x}{\partial h / \partial y}(x^*)$$

Step 2: FOC written in terms of 1 variable x , evaluated at x^*

$$F'(x^*) = \frac{\partial f}{\partial x}(x^*) + \frac{\partial f}{\partial y}\phi'(x^*) = 0$$

Expanding (simplifying (x^*, y^*))

$$\begin{aligned} F'(x) &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\phi'(x) + \lambda \left[\frac{\partial h}{\partial x} + \frac{\partial h}{\partial y}\phi'(x) \right] \\ &= \frac{\partial f}{\partial x} + \lambda \frac{\partial h}{\partial x} + \frac{\partial f}{\partial y}\phi'(x) + \lambda \frac{\partial h}{\partial y}\phi'(x) \\ &= \frac{\partial \mathcal{L}}{\partial x}(x, \phi(x)) + \phi'(x) \frac{\partial \mathcal{L}}{\partial y}(x, \phi(x)) \end{aligned}$$

Step 3: Compute the second derivative

$$\begin{aligned}
 F''(x^*) &= \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial \mathcal{L}}{\partial x}(x^*, y^*) + \phi'(x^*) \frac{\partial \mathcal{L}}{\partial y}(x^*, y^*) \right) \\
 &= \frac{\partial^2 \mathcal{L}}{\partial x^2} + \frac{\partial^2 \mathcal{L}}{\partial x \partial y} \frac{\partial y}{\partial x} + \phi'(x) \left[\frac{\partial^2 \mathcal{L}}{\partial y \partial x} + \frac{\partial^2 \mathcal{L}}{\partial y^2} \frac{\partial y}{\partial x} \right] + \frac{\partial \mathcal{L}}{\partial y} \phi''(x) \\
 &= \frac{\partial^2 \mathcal{L}}{\partial x^2} + 2 \frac{\partial^2 \mathcal{L}}{\partial x \partial y} \phi'(x^*) + \frac{\partial^2 \mathcal{L}}{\partial y^2} \phi'(x^*)^2 \quad \left(\frac{\partial \mathcal{L}}{\partial y} = 0 \text{ by FOC} \right) \\
 &= \frac{\partial^2 \mathcal{L}}{\partial x^2} + 2 \frac{\partial^2 \mathcal{L}}{\partial x \partial y} \left(-\frac{\partial h / \partial x}{\partial h / \partial y} \right) + \frac{\partial^2 \mathcal{L}}{\partial y^2} \left(-\frac{\partial h / \partial x}{\partial h / \partial y} \right)^2 \\
 &= \frac{1}{\left(\frac{\partial h}{\partial y} \right)^2} \left[\frac{\partial^2 \mathcal{L}}{\partial x^2} \left(\frac{\partial h}{\partial y} \right)^2 - 2 \frac{\partial^2 \mathcal{L}}{\partial x \partial y} \frac{\partial h}{\partial x} \frac{\partial h}{\partial y} + \frac{\partial^2 \mathcal{L}}{\partial y^2} \left(\frac{\partial h}{\partial x} \right)^2 \right] \\
 &= \frac{1}{\left(\frac{\partial h}{\partial y} \right)^2} (-|H|) \tag{8}
 \end{aligned}$$

Why does (8) hold? Compute det of the bordered Hessian:

$$\begin{aligned}
 |H| &= 0 \begin{vmatrix} \frac{\partial^2 \mathcal{L}}{\partial x^2} & \frac{\partial^2 \mathcal{L}}{\partial x \partial y} \\ \frac{\partial^2 \mathcal{L}}{\partial y \partial x} & \frac{\partial^2 \mathcal{L}}{\partial y^2} \end{vmatrix} - \frac{\partial h}{\partial x} \begin{vmatrix} \frac{\partial h}{\partial x} & \frac{\partial^2 \mathcal{L}}{\partial x \partial y} \\ \frac{\partial h}{\partial y} & \frac{\partial^2 \mathcal{L}}{\partial y^2} \end{vmatrix} + \frac{\partial h}{\partial y} \begin{vmatrix} \frac{\partial h}{\partial x} & \frac{\partial^2 \mathcal{L}}{\partial x^2} \\ \frac{\partial h}{\partial y} & \frac{\partial^2 \mathcal{L}}{\partial y \partial x} \end{vmatrix} \\
 &= -\frac{\partial h}{\partial x} \left(\frac{\partial h}{\partial x} \frac{\partial^2 \mathcal{L}}{\partial y^2} - \frac{\partial h}{\partial y} \frac{\partial^2 \mathcal{L}}{\partial x \partial y} \right) + \frac{\partial h}{\partial y} \left(\frac{\partial h}{\partial x} \frac{\partial^2 \mathcal{L}}{\partial y \partial x} - \frac{\partial h}{\partial y} \frac{\partial^2 \mathcal{L}}{\partial x^2} \right) \\
 &= -\left(\frac{\partial h}{\partial x} \right)^2 \frac{\partial^2 \mathcal{L}}{\partial y^2} + 2 \frac{\partial^2 \mathcal{L}}{\partial x \partial y} \frac{\partial h}{\partial x} \frac{\partial h}{\partial y} - \left(\frac{\partial h}{\partial y} \right)^2 \frac{\partial^2 \mathcal{L}}{\partial x^2},
 \end{aligned}$$

Conclusion

- ① If $|H| > 0$, then (8) is negative, the critical point is a MAX.
- ② If $|H| < 0$, then (8) is positive, the critical point is a MIN.

Problem setup:

$$\max(\min) f(x, y) \quad \text{s.t.} \quad g(x, y) = c$$

FOC

$$\frac{\partial \mathcal{L}}{\partial x_1} = f_x - \lambda g_x = 0 \iff f_x = \lambda g_x,$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = f_y - \lambda g_y = 0 \iff f_y = \lambda g_y$$

Let the optimal point satisfying FOC be $f^*(c)$. Total derivative of $f^*(c)$ with respect to c , we obtain:

$$\begin{aligned} \frac{df^*(c)}{dc} &= f_x(x^*, y^*) \frac{dx^*}{dc} + f_y(x^*, y^*) \frac{dy^*}{dc} \\ &= \lambda(c) g_x(x^*, y^*) \frac{dx^*}{dc} + \lambda(c) g_y(x^*, y^*) \frac{dy^*}{dc} \\ &= \lambda(c) \underbrace{(g_x(x^*, y^*) dx^* + g_y(x^*, y^*) dy^*)}_{dg(x^*, y^*)} / dc \end{aligned}$$

By definition $g(x, y) = c$, so $dg = dc$. It follows that

$$\frac{df^*(c)}{dc} = \lambda(c)$$

Assuming dc is small, near 0, then we have

$$f^*(c + dc) - f^*(c) \approx \lambda(c)dc$$